QUANT

Time, Speed & Distance

eBOOK

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INDEX

Time, Speed &Distance E-Book

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Time, Speed & Distance

1.1 Basics

Time, speed and distance (TSD) is one of the most popular topics in CAT. Almost 2 to 3 questions in the paper are asked from this chapter every year. The concept of time, speed and distance is also used extensively for questions relating to different areas in various other examinations.Students should try to understand the inter-relationship between the factors time, speed and distance.

The most important relationship between time, speed and distance is

Distance Travelled = Speed × Time

And the different ways of expressing the same relationship will also come in handy while solving problems in this section. They are

Speed = Distance ÷ Time and

Time = Distance ÷ Speed

The problems that occur in this topic can be classified into one of the following sub headings:

(a) **Speed, Time & Distance**

(b) **Trains**

(c) **Relative Speed**

(d) **Races**

1.2 Important Distance and Time Conversions

1 km = 1000 meter

1 meter = 100 cm

1 hour = 60 min

1 min = 60 sec

1 hour = 3600 sec

1 km/hr **=** $\left(\frac{1\times1000}{1\times3600}\right)=\frac{5}{18}$ **m/sec**.

1 mile = 1760 yards

1 yard = 3 feet

1 mile = 5280 feet

1 mph **=** $\left(\frac{1\times1760}{1\times3600}\right)=\frac{22}{45}$ yards/sec

1 mph **=** $\left(\frac{1\times5280}{1\times3600}\right)=\frac{22}{15}$ ft/sec

1.3 Solved examples on Speed, Time and Distance

Usually, problems in this subheading cover finding one of the values, when the other two are provided. The important thing to note in such problems is that the units of all three quantities used are the same, i.e. if speed in km/hr, then distance will have to be in km and time in hours.

Ex.1 What is the distance covered by a car traveling at a speed of 40 kmph in 15 minutes?

Sol: $40 \times \frac{15}{60}$ = 10 km.

The important point to note is that time given was in minutes, whereas the speed was in kmph. Therefore, either speed will have to be expressed as km/min or time will have to be expressed in hours to apply the relationship.

In this case we converted time into hours to get the answer. Conversely, converting speed into km/min, we get 40 kmph = $^{40}/_{60}$ km/min = $^{2}/_{3}$ km/min

Therefore, distance traveled = $15 \times \frac{2}{3}$ = 10 km.

Ex.2 Traveling at a speed of 50 Kmph, how long is it going to take to travel 60 km?

Sol: Distance = Speed $\times$ Time $\because$ Time = Distance $\div$ Speed $\Rightarrow \frac{60}{50}$ = 1.2 hours = 1 hour and 12 minutes.

Note: While converting decimal hours into the minutes, these are to multiplied with 60 and not by hundred.

Ex.3 A man covers 75 km in 90 minutes. What is his speed in km/h?

Sol: Speed = Distance $\div$ Time.

Since, time is given in minutes and the required answer is in km/h, we need to convert time into equivalent hours.

90 minutes = $\frac{90}{60}$ = 1.5 hours.

Therefore, speed = $\frac{75}{1.5}$ = 50 km/h.

Ex.4 Walking $^{5}/_{6}{}^{th}$ of his usual speed, Mike reached his destination 10 minutes late. Find his usual time, and the time taken on this occasion?

Sol: Let his usual speed be *x* km/hr and his usual time be *t* hours.

His time on this occasion is $\frac{5}{6}x$,

The time taken is $\left(t + \frac{10}{60}\right)$ hours.

Since the distance traveled on both occasions is the same, $xt = \frac{5x}{6} \times \left(t + \frac{10}{60}\right)$.

Solving for *t*, we get $t = \frac{5}{6}$ hours

= 50 minutes, and the time taken on this occasion

= 50 + 10 = 60 minutes.

Ex.4(a) If the distance traveled by Mike be 60 km, then what was his usual speed and what was the speed on this occasion?

Sol: Usual time taken = 50 minutes = $\frac{5}{6}$ hours

The distance = 60 km.

Usual Speed = Distance÷Usual Time$\Rightarrow \frac{60}{5/6}$ = 72 kmph.

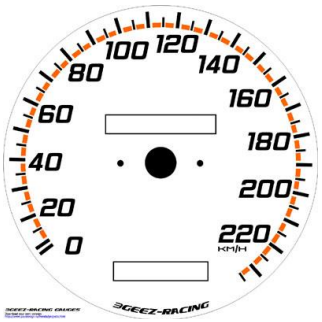
Speed on this occasion =

Distance ÷ Time on this occasion = $\frac{60}{1}$= 60 kmph.

The ratio between the usual speed to the speed on this occasion = $\frac{72}{60} = \frac{6}{5}$

The ratio of the usual time taken to the time taken on this occasion = $\frac{50.\text{min}}{60.\text{min}} = \frac{5}{6}$.

<u>Note</u>: In general, **<u>speed and time have an inverse relationship</u>**. Therefore, if the speed becomes, say 0.5 times the original speed, then the time taken becomes twice as much as taken originally for the same distance. Or if the ratio of the speed of two moving objects is in the ratio of 3:4, the time taken by them to cover identical distance will be in the ratio of 4:3.



1.4 Average Speed

Case 1: When the time taken is the same:

The average speed of traveling at two different speeds for the same time span is just the simple average of these two speeds. So, if two speeds are a and b units.

$$\textbf{Average Speed} = \frac{a+b}{2}$$

Case 2: When the distance travelled is the same:

However, the above simple average rule does not work when the time span of each of the different speeds is different and only the distance is the same. In this case, one should take the simple average of the inverses of the two speeds and then again inverse the speed.
Let the two speeds be *a*kmph and *b*kmph.
Let the distance traveled in each of the speeds be *x* km.

Time taken to cover *x* km at '*a*' kmph= $\frac{x}{a}$ & at '*b*' kmph= $\frac{x}{b}$.

Total time taken = $\frac{x}{a} + \frac{x}{b} = \frac{bx+ax}{ab} = \frac{x(b+a)}{ab}$, and the total distance covered = 2*x*.

Therefore, average speed = $\dfrac{2x}{x(a+b)/ab} = \dfrac{2ab}{a+b}$.

The average speed of traveling at two different speeds when the distance travelled is same, is harmonic mean of these two speeds. So, if two speeds are a and b units.

$$\textbf{Average Speed} = \frac{2ab}{a+b}$$

1.5 Solved examples on Average speed

Ex.5 A motorist travels one hour at an average speed of 45 kmph and the next hour at an average speed of 65 kmph. Then his average speed is

Sol: (45 + 65) ÷ 2 = 55 kmph.
The total distance traveled by the motorist in these two hours = 65 + 45 = 110 km and he has taken two hours. Therefore, his average speed = 55 kmph.

Ex.6 On my way from the office to the Pimpri class, I drive at 30 kmph and on the return journey I drive at 45 kmph. What is my average speed of travel?

Sol: 37.5 kmph is incorrect as the time traveled is different in both the cases and only the distances are same.
Let the distance between the office and Pimpri class be *x* km.

∴ Time taken on my onward journey = $\dfrac{x}{30}$ hours and

Time taken on my return journey = $\dfrac{x}{45}$.

∴ The total time taken for my onward and return journey

$= \dfrac{x}{30} + \dfrac{x}{45} = \dfrac{5x}{90}$ hours.

The total distance traveled both ways = 2*x* km

∴ Average speed = $\dfrac{2x}{5x/90}$ = 36 kmph.

1.6 Relative Speed and Trains

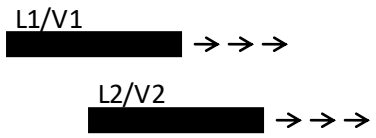
- When two objects are moving in the same direction, then their relative speed is the difference between the two speeds.
- When two objects are moving in the opposite direction, then their relative speed is the sum of the two speeds.
- Problems in this section will involve finding the distance of a train
- When it crosses a stationary man / lamp post / sign post / pole - in all these cases the object which the train crosses is stationary - and the distance traveled is the length of the train.
- When it crosses a platform / bridge - in these cases, the object which the train crosses is stationary - and the distance traveled is the length of the train + length of the object
- When it crosses another train which is moving at a particular speed in the same / opposite direction - in these cases, the other train is also moving and the relative speed between them is taken depending upon the direction of the other train - and the distance is the sum of the lengths of both the trains
- When it crosses a car / bicycle / a mobile man - in these cases again the relative speed between the train and the object is taken depending upon the direction of the movement of the other object relative to the train - and the distance traveled is the length of the train.

Let us see some cases:

Case 1. **When two trains are crossing each other**

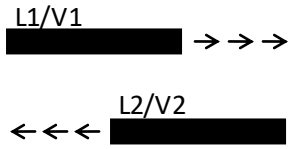
Let us assume that the trains of length L1 and L2 and speed V1 m/s and V2 m/s respectively are crossing each other;

i. The direction of the movement of both the trains are the same:



Relative speed = | V1 - V2 |
Total distance covered = L1 + L2

ii. The direction of the movement of both the trains are opposite:



Relative speed = | V1 + V2 |
Total distance covered = L1 + L2

Case 2. **When a train is crossing a stationary object**

i. When the trains is crossing a pole or a stationary human being:

(Fig. 1)

(Fig. 2)

Let us assume that A is a pole. In figure I, the front of the train is about to cross the pole and in figure 2, the tail of the train has just crossed the pole. It is understood here that the train has crossed its whole length with respect to the pole. So, when the train is crossing any stationary object of negligible width, total distance covered is its own length.

Relative speed = V1 + V2 since V2 = 0, then, the relative speed = V2
Total distance covered = L1 + L2, since L2 (width of the pole) is negligible with respect to L2 (Length of the train), so we do not consider h while calculating the quantities. Thus, distance = L1.

However, it should be remembered that this is mathematically not correct and all the solutions are on the assumption that the width of the pole is zero, which is obviously not true.

Case 3. **When the train is crossing a platform or a standing train.**

Relative speed = V1 + V2
Where V1 is the speed of the moving train and V2 is the speed of the standing train or the platform.
Since V2 = 0, so the relative speed = V1
Total distance covered = L1 + L2
Where L1 is the length of the moving train and L2 is the length of the standing train or the platform.

Ex.7 A train traveling at 60 kmph crosses a man in 6 seconds. What is the length of the train?

Sol: Speed in m/sec = 60 × 5/18= 50/3 m/sec.
Time taken to cross the man = 6 seconds.
Therefore, distance traveled = 50/3 × 6 = 100 meter
= length of the train.

Ex.8 A train traveling at 60 kmph crosses another train traveling in the same direction at 50 kmph in 30 seconds. What is the combined length of both the trains?

Sol: Speed of train A in m/sec = 60 × 5/18= 50/3m/sec
Speed of train B in m/sec = 125/9 m/sec
The relative speed is = 50/3 - 125/9= 25/9 m/sec.
Time taken for train A to cross train B = 30 seconds.
Therefore, distance traveled = 25/9×30 = 250/3 meter
= Combined length of two trains.

1.7 Boats and Streams

Downstream movement:When the direction of the movement of a river and a boat is the same, their collective movement is known as the downstream movement. And the distance covered by boat is known as downstream distance.

Upstream movement:When the direction the movement of the river and a boat is opposite, they are said to be in upstream movement. The distance covered in this case is known as upstream distance. If the speed of the river = R and the speed of the boat - B. then upstream speed = B - R (Conventionally the speed ofboat is taken more the than speed of the river.)

In most of the cases of boats and streams. The distance covered downstream and upstream are the same.In those cases, the ratio of the time taken becomes inverse of the ratio of the speeds.

i.e. $\dfrac{\text{Time taken downstream}}{\text{Time taken upstream}} = \dfrac{\text{upstream speed}}{\text{downstram speed}}$

So we can say if the speed of the boat in still water is say *B*kmph and if the speed at which the stream is flowing is *W*kmph, then

(i) When the boat is traveling with the stream the speed of the boat = (*B* + *W*) kmph
(ii) When the boat is traveling against the stream the speed of the boat = (*B* – *W*) kmph.
(iii) If the upstream is denoted as *U* and downstream is denoted as *D* then
(iv) $B = \dfrac{D+U}{2}$., $W = \dfrac{D-U}{2}$.
Hence, if downstream speed and upstream speed are given as 20 km/h and 10 km/h respectively, then the speed of the boat = 15 km/h and speed of the river = 5 km/h.

1.8 Solved Examples on Boats and Streams

Ex.9 A boat travels at a speed of 30 kmph with the stream and 18 kmph against the stream. What is the boat's speed in still water and the speed of the stream?

Sol: Speed of the boat in still water = $\dfrac{30+18}{2}$ = 24 kmph and Speed of the stream = $\dfrac{30-18}{2}$ = 6 kmph

Ex.10 Driving 5/4th of his usual speed, David reached the destination 12 minutes earlier. What is the usual time he takes to travel?

Sol: Let *X* km/hr be the usual speed and let t hours be the usual time taken.

Speed on this occasion = $\dfrac{5}{4}X$ km/hr

The time taken on this occasion = $\left(t - \dfrac{12}{60}\right)$ hrs.

Since the distance is the same in both the cases, $Xt = \dfrac{5}{4}X \times \left(t - \dfrac{12}{60}\right)$

Solving for *t*, *t* = 1 hour - the usual time taken.

Ex.11 In a cross-country race, a motorist averages a speed of 140 mph during the first 4 hours and then increases his average by 20 mph during the last 3 hours. What was his average speed during the entire race?

Sol: Distance traveled in first 4 hours = 140 $\times$4 = 560 miles
Distance traveled in next 3 hours = 160 $\times$ 3 = 480 miles
Therefore, the total distance traveled = 1040 miles.
The total time taken = 7 hours.
Therefore, the average speed = $\frac{1040}{7}$ = $148\frac{4}{7}$ mph.

Ex.12 During the onward journey from Bombay to Pune, Deccan Queen travels at an average speed of 80 kmph, while on the return journey, the train is able to average a speed of 100 kmph. What is the average speed of the train on its entire journey?

Sol: Average speed = $\frac{2ab}{a+b} = \frac{2\times 80\times 100}{180}$
= $\frac{800}{9}$ = $88\frac{8}{9}$ km/hr

Ex.13 A train traveling at 100 km/hr crosses a bridge of half a km length completely in 30 seconds. What is the length of the train?

Sol: Speed = 100 km/hr = 100 $\times\frac{5}{18}$ = $\frac{250}{9}$ m/sec
Time taken to cross = 30 seconds.
Therefore, distance traveled = $\frac{250}{9}\times$ 30 = $\frac{2500}{3}$ m
Distance = Length of the train + length of the bridge
$\frac{2500}{3}$ = Length of the train + 500
$\Rightarrow$ Length of the train = $\frac{1000}{3}$ m

Ex.14 A train crosses a signpost in 6 seconds and a car traveling in the same direction at 50 kmph in 72 seconds. What is the length of train and the speed at which it is traveling?

Sol: *Case I:*
Let X km/hrbe the speed of the train. = $X\times\frac{5}{18}$ m/sec
Time taken to cross a signpost = 6 seconds.
Therefore distance traveled = $X\times\frac{5}{18}\times 6 = \frac{5X}{3}$ meter
= length of the train

Case II:
The speed of the car = 50 km/hr.
Relative speed of the train wrt car = (X – 50) km/hr
= (X – 50) $\times\frac{5}{18}$ m/sec.
Time taken to cross the car = 72 seconds.
Therefore, distance traveled = (X – 50) $\times\frac{5}{18}\times$ 72
= 20 (X-50) m = length of train
Equating length of the train in Case I and Case II,

we get $\frac{5X}{3}$ = 20 (X – 50).

Solving for X, we get $X = \frac{600}{11}$ km/hr

and the length = $\frac{5}{3} \times \frac{600}{11} = \frac{1000}{11}$ m

Ex.15 Anvolvo bus and an ordinary bus leave Pune for Chinchwad - a distance of 32 km simultaneously. The ratio between the average speed of the volvo bus and that of the ordinary bus is 3:2. The volvo bus reaches Chinchwad and immediately leaves back for Pune and meets the ordinary bus at Pimpri. What is the distance between Chinchwad and Pimpri?

Sol: Speed of volvo : Speed of ordinary :: 3 : 2
Since, Distance $\propto$ Speed,
Distance traveled by volvo : Distance traveled by ordinary :: 3 : 2
Let distance between Chinchwad and Pimpri be x km.
Then distance traveled by volvo = 32 + x, while the distance traveled by ordinary bus = 32 – x.
Therefore, 32 + x : 32 – x :: 3 : 2.
Solving for x, we get x = 6.4 km
= distance between Pimpri and Chinchwad.

Ex.16 The ratio of the speed at which Samar can row with the stream and against the stream is 5 : 3. If he takes 2 hours to row with the stream on his onward journey of 90 km, what is his rowing speed in still water and the speed of the river?

Sol: Speed with stream : Speed against stream :: 5 : 3
Distance = 90 km. Time taken with the stream = 2 hrs.

Speed with the stream = $\frac{90}{2}$ km/hr = 45 km/hr.

So speed against the stream = 45 $\times \frac{3}{5}$ = 27 km/hr.

Hence speed in still water = $\frac{27+45}{2}$ = 36 km/hr and

The speed of the stream = $\frac{45-27}{2}$ = 9 km/hr.

Ex.17 Traveling at 6 km/hr, I reach my office 20 minutes late. Traveling at 8 km/hr I reach my office 30 minutes early. What is my usual speed and time taken to reach my office?

Sol: Let my usual speed be S km/hr and my usual time be t hours.

Therefore, $6 \times \left(t + \frac{20}{60}\right) = 8 \times \left(t - \frac{30}{60}\right)$. Solving for t, we get t = 3 hours.

Since the usual time taken = 3 hours, usual distance traveled = 3S kms,
Equating distance traveled usually, with distance traveled at any of the other two speeds, we get

$6 \times \left(3 + \frac{20}{60}\right) =$ 3S. Therefore, S = $6\frac{2}{3}$ km/hr.

Ex.18 Sajan hikes up a hill at 4 mph and comes down at 6 mph. If the total time taken for the total journey is 3.5 hours, what was the distance between the hilltop and the foothills?

Sol: Average speed = $\frac{2ab}{a+b} = \frac{2 \times 6 \times 4}{10}$ = 4.8 mph
Time taken = 3.5 hours both ways.
So, the two way distance = 4.8 $\times$ 3.5 miles = 16.8 miles.
Hence, the distance one-way = 8.4 miles.

Ex.19 Siya leaves Pune for Bombay at 17:30 hours and reaches Bombay at 21:30 hours, while Shatabdi, which leaves Bombay at 17:00 hours, reaches Pune at 20:30 hours. At what time do they pass each other?

Sol: Let the distance between Bombay and Pune = *d* km.

Siya Speed = $\frac{d}{4}$ km/hr& that of Shatabdi = $\frac{d}{3.5}$ km/hr.

Let t be the time in hrs after Shatabdi has left for Pune, when the two trains meet.

Therefore, distance traveled by Shatabdi = $\frac{d}{3.5} \times t$.

And that of Siya= $\frac{d}{4} \times \left(t - \frac{30}{60}\right)$.

The sum of the distances traveled by the two trains
= distance between Bombay and Pune = *d* km.

Therefore, $\frac{d}{3.5} \times t + \frac{d}{4} \times \left(t - \frac{30}{60}\right) = d$.

Solving for t, we get t = 2.1 hours = 2 hrs 6 min.
Hence, the two trains meet at 7:06 pm.

1.9 Concept of Circular Motion

Let us explore the concept of circular motion. It involves analyzing problems which involve circular tracks. people generally move around a track clockwise or anti clockwise at different speeds. The basic objective of this concept is generally to calculate:

I. the time of meeting of people (running around the track) at the starting point again after they started.
II. meeting for the first time anywhere on the track.
III. At how many distinct points do people meet while running on the circular track.

Time to meet at the Starting point: The concept here is that firstly the time taken by each of the persons to run one complete round at their respective speeds and then take the L.C.M of these times. This LCM gives the time of their meeting again at the starting point.

Example: There is a track with a length of 120 mtrs and 2 people, A & B, are running around it at 12 m/min and 20 m/min respectively in the same direction.

Now, the time of their meeting again at the starting point will be the LCM of $\frac{120}{12}$ & $\frac{120}{20}$ i.e. 10 & 6, which is 30 mins. So, after 30 mins these people will be together at the starting point.

You can also check this as after 30 mins A would have taken 3 rounds and B would have taken $\frac{30}{6} = 5$ rounds. So after completing 3 & 5 rounds they will be at the starting point.

Time to meet anywhere: Now to meet for the first time the faster person has to complete one full round extra over the slower person. The faster person is ahead of the slower one right from the first minute only due to his speed being higher than the speed of the other and they both are moving in the same direction. It can be said that when the faster is ahead of the slower by one full track length, he will be overtaking the slower person from behind. Now, at this very moment these people meet.

In order to calculate the time we can say that time of meeting = $\frac{\text{track length}}{\text{relative speed}}$.

Taking the previous data example's data to find when will B overtake A,

Time of meeting = $\frac{120}{20-12} = \frac{120}{8} = 15\,\text{min}$.

In order to visualize we can say that B covers 8 mtr/min extra over A. So when B covers 120 mtrs. extra he will overtake A from the behind and hence they both will meet.

No. of points: Let's suppose the question now is at how many different points do people running around a circular track meet.

The logic that operates behind this problem: if we divide the time of their first meeting at the starting point with the time of their first meeting anywhere on the track, we get the number of points at which these people would meet including the starting point.

Again considering the data from the above example, we can say that number of points = $\frac{30}{15} = 2$ points

This is inclusive of the starting point. There are two points, where they will meet.

Shortcut to find the number of distinct meeting points:

Step 1: Take the lowest ratio of the speeds.
Step 2: Add if objects are moving in opposite directions.
Step 3: Subtract if objects are moving in same directions.

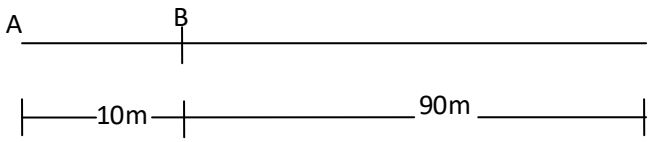
So, all the three parts can be represented in the following table:

	Same Direction	**Opposite Direction**
Time to reach initial point first time	L.C.M $\left\{\frac{x}{a}, \frac{x}{b}\right\}$	L.C.M $\left\{\frac{x}{a}, \frac{x}{b}\right\}$
Time to meet for the first time on track	$\frac{x}{\lvert a-b \rvert}$	$\frac{x}{a+b}$
No. of distinct meeting Points	Lowest ratio of a & b =c:d (say) Then No. of distinct meeting Points= |c-d|	Lowest ratio of a & b =c:d (say) Then No. of distinct meeting Points= c+d

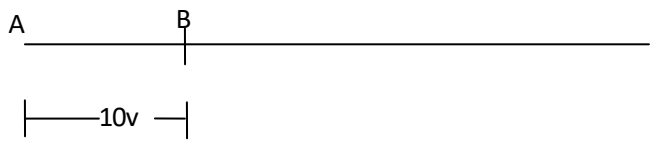
1.10 Races

In a race, one can give other a head start of distance or time. When the terminologies, A gives B a start of 10 meter or A beats B by 10 meter in a 100 meter race, is used, it means that when A completes 100 meter B would have completed only 90 meter. So, we can take following two cases.

Case 1. A gives a start of 10m to B → When B has already run 10m, then A starts running.



Case 2. A gives a start of 10 secs to B →B has already run for 10 secs, now A starts running.



Where V m/s is the speed of B.

Ex.20 In a 400 meter race, A can give B a start of 25 meter and C a start of 50 meter. What start can B give C?

Sol: When A runs 400 meter, B runs 375 meter.
When A runs 400 meter, C runs 350 meter.
i.e. when B runs 375 meter, C runs 350 meter.
Therefore, when B runs 400 meter,
C runs $\frac{350 \times 400}{375}$ = 373.33 meter.
Or B can give C a start of 26.67 meter in a 400 meter race.

Ex.21 A and B start from the same point to run in opposite directions on a circular ground of 1 mile circumference. A gives B a start of 160 yards and meets him after A has covered 880 yards. When A reaches the starting point how far would he be from B (given 1 mile = 1760 yards)?

Sol: Circumference of the ground = 1 mile = 1760 yards.
A gives B a start of 160 yards - A starts after B has covered 160 yards. When A has covered 880 yards, he meets B, by which time B has also covered 880 yards as the circumference of the ground = 1760 yards.
When A covers 880 yards, B covers (880 - 160) yards i.e. 720 yards.i.e. their speeds are in the ratio of 880 : 720
= 11 : 9
Therefore, when A reaches the starting point - i.e. another 880 yards, B would have covered 720 yards in the opposite direction and would have covered a total of
160 + 720 + 720 = 1600, which is 160 yards away from the starting point and A

Ex.22 A gives B a start of 20 meter in a 200-meter race and B gives C a start of 27 meter in a 300 meter race. How much start can A give C in a half km race?

Sol: Race 1: Length 200 meter.
When A reaches 200 meter, B reaches 180 meter.
⇒When A reaches 100 meter B reaches 90 meter.
Race 2: Length 300 meter.
When B reaches 300 meter, C reaches 273 meter
⇒When B reaches 100 meter, C reaches 91 meter.

Therefore, in a 100 meter race, when A finishes 100 meter C finishes $\frac{90 \times 91}{100}$ = 81.9 meter.

In other words, A can give C a start of 18.9 meter in a 100-meter race.
Therefore, in a half a kilometre race, A can give C a start of 90.5 meter.

Ex.23 A and B start jogging in a circular park of circumference 150 metre from the same point simultaneously in the same direction. A jogs at a speed of 36 kmph and B jogs at a speed of 90 kmph. When will they meet ever again and when will they meet at the starting point?

Sol: Circumference of the park =150 metre.
Speed of A = 36 kmph = 36 × 5/18 = 10 m/sec
Speed of B = 90 kmph = 90 × 5/18 = 25 m/sec
They will meet again first time when B has covered one full circle more than A. i.e. when their relative speeds completed one full circle.
Their relative speed = 25 – 10 = 15 m/sec.
Time taken to complete 1 full circle at the speed of 15 metre/sec = 150/15 = 10 sec.Thus they meet after 10 seconds. They will meet again at the starting point at a time, which is the LCM of the time taken by each to complete 1 circle.
Time taken by A to complete a round = 150/10 = 15 sec.
Time taken by B to complete a round = 150/25 = 6 sec.
LCM of 15 and 6 = 30 seconds. At that time, A would have completed 2 full rounds and B would have completed 5 full rounds.

Here you could have also taken LCM in fractions only i.e. LCM of $\frac{150}{10}$ *and* $\frac{150}{25} \Rightarrow$ 150/5 = 30 seconds.

Ex.24 Tom can run a km in 9 minutes and 54 seconds, while Jerry can run in 10 minutes. How many metres start can Tom give Jerry?

Sol: Since, Jerry takes 6 seconds longer than Tom, the start that Tom can give Jerry is the distance that Jerry can cover in 6 s.
In 10 minutes Jerry can run 1000 metre.
Therefore, in 6 seconds he can run 1000 × 6/600 = 10 metre

Ex.25 In a km race, A gives B and C starts of 6 meter and 12 meter respectively. When A crosses the finishing line, B is 4 meter behind him and C is 18 meter behind him. How much start can B give C in a km race so that they finish together?

Sol: When A completes 1000 m, B completes 1000 – 6 – 4 = 990 m.
When A completes 1000 m, B completes 1000 – 12 – 18 = 970 m.
i.e. when B completes 990 m, C completes 970 m.

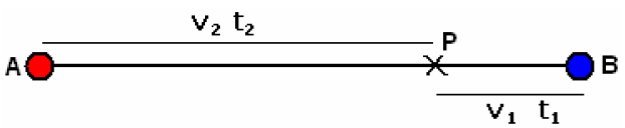
Therefore, when B completes 1000 meter C completes $\frac{1000 \times 970}{990} = \frac{97000}{97} m$

Therefore, the start that B can give C is

1000 - $\frac{97000}{99} = \frac{2000}{99} = 20\frac{20}{99}$ meter.

1.11 One Special Case

Crossing of Two bodies



Let two bodies start from A and B simultaneously towards each other with speeds v_1 and v_2, respectively. After meeting at point P, they take time t_1 and t_2 to reach their destinations.
The distance AP = v_2t_2 and distance PB = v_1t_1.

Now since the bodies start simultaneously, they take the same time to meet at P.

$$\Rightarrow \frac{AP}{v_1} = \frac{PB}{v_2} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{t_2}{t_1}}.$$

This is a special case to remember.

Ex.26 One day I started from A to B at exactly 12 noon. My friend started from B to A at exactly 2:00pm. We met on the way at five past four reached our destinations at exactly the same time. What time was it?

Sol: I have traveled for 245 minutes whereas my friend has traveled for 125 minutes when we meet. If we reverse the direction of travel, both my friend and I started at the same time, crossed each other and took 125 minutes to reach our destinations. Therefore, $\frac{v_1}{v_2} = \sqrt{\frac{125}{245}} = \frac{5}{7}$. Therefore, the distance that I traveled in 245 minutes, my friend will travel in 245 × 5/7 = 175 minutes. Therefore, total time of travel for my friend = 125 + 175 = 300 minutes. Therefore, we reach our destinations at 7 : 00 pm.

Ex.27 A motorboat can go 10 miles downstream on a river in 20 mins. It takes 30 mins for this boat to go back at the same 10 miles. Find the speed of the current.

Sol: as the distance traveled is constant, the time taken is inversely proportional to speed.

$$\Rightarrow \frac{v+u}{v-u} = \frac{30}{20} \Rightarrow \frac{v}{u} = 5 \Rightarrow v + u = 6u = 10/20/60 = 30m/h \Rightarrow u = 5m/h.$$

1.12 Escalators

Escalators are simple application of upstream' and downstream' concept with a twist-the distance to be traveled, i.e. the total length of the escalator (measured in the number of stairs) remains constant.

If you think about it, a person will climb more stairs than the total length if he's climbing against the escalator movement and he'll climb less stairs than the total length if he's climbing in the same direction as the escalator movement. Let a person's climbin speed be v stairs/minute and the escalator speed be u stairs/minute. Let the person climb N stairs in total to cover the escalator. To climb N stairs hell take time t = N/v minutes. In this time, escalator will throw out (N/v) × u stairs.

Case 1: Climbing against the escalator– Total length of escalator = Number of stairs climbed – stairs combing out of the escalator in the same time = N – (N/v) × u

Case 2: Climbing with the escalator– Total length of escalator = Number of stairs climbed + stairs coming out of the escalator in the same time = N + (N/v) × u

Ex.28 I am in the habit of waling up the subway escalator while it is running. I climb 20 steps at my normal pace and it takes me 60 seconds to reach the top, whereas my wife climbs only 16 steps and it takes her 72 seconds to reach the top. If the escalator broke down tomorrow, how many steps would I have to climb?

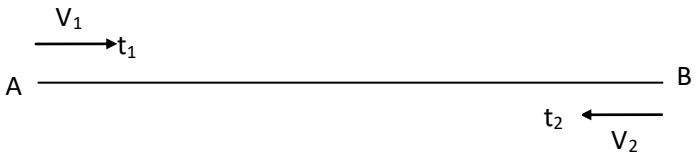
Sol: Let the speed of the escalator be u stairs/sec. Since the length of the escalator is constant 20 + 60 u = 16 + 72u ⇒ u = 1/3 stairs/ sec. therefore, the total length of the escalator = 20 + 60u = 40 stairs.

Use of Arithmetic and Harmonic Progression in Time, Speed and Distance

Here is a textbook situation in Time, Speed and Distance: A man goes from point A to point B with velocity v_1 and returns with velocity v_2. What is his average velocity?

Using the formula $\text{Average Velocity} = \frac{\text{Total Distance}}{\text{Total Time}}$, the average velocity can be found to be $V_{average} = \frac{2v_1v_2}{v_1 + v_2}$. So far so good. Ever wondered why we get this result? Why do we get the velocity as the harmonic mean of the two velocities? The answer lies in basics of arithmetic and harmonic progressions.

Let t_1 and t_2 be the taken while going from A to B and coming back. The situation is shown below:



Now what will a hypothetical average velocity in this situation mean? It will mean that a person takes the same time, $t_{average,}$ while going from A to B and coming back. The situation is summarized below:

The total time taken will be same as $t_1 + t_2$ or $2 \times t_{average} = t_1 + t_2$

In another words, t_1, $t_{average}$, and t_2 will be in arithmetic progression.

So how is this related to velocity?

Remember that when distance is constant velocity is inversely proportional to time?

i.e. V is proportional to 1/T or V = k/T.

If T_1, T_2, and T_3 were in **arithmetic progression**, then $1/T_1$, $1/T_2$, $1/T_3$ are in **harmonic progression**⇒ k/T_1, k/T_2, k/T_3, are in **harmonic progression**⇒ V_1, V_2 and V_3 are in **harmonic progression!**

It can also be proved that if V_1, V_2 and V_3 are in **arithmetic progression**, then T_1, T_2, and T_3 are in **harmonic progression.**

1.13 Clocks

A clock is a typical example of a circular motion. Two persons named minute hand and hour hand are moving in same (clockwise) direction.

While solving the problem of clocks, we must remember following points:

- The distance is measured in degrees. There are 30 degrees between any two consecutive hour positions. So, the total track length is considered as 360° . At 4:00 o-clock, the distance between the minute hand and the hour hand is 120°.
- The speed is measured in degrees per minute. The minute hand rotates one round in one hour, or it travels 360° in 60 minutes. Therefore, the speed of the minute hand is 6°/min. The hour hand travels 30° (from one hour position to the next) in one hour. Therefore, the speed of the hour hand is 1/2°/min.
- The relative speed of the minute hand with respect to the hour hand is 6° per min – 1/2° per min = 11/2° per min. Therefore, the minute hand chases the hour hand around the circle with a relative speed of 11/2° per min.

Ex.29 At what time between 4:00 pm and 5:00pm, will the minute hand and the hour hand meet?

Sol: At 4:00pm, the distance between the minute hand and the hour hand is 120°.

Therefore, the minute hand is 120° behind at 4:00pm and then it starts closing the gap by 11/2° every minute. Therefore, the time taken to catch up with hour hand completely

$= \frac{\text{Initial gap}}{\text{Distance reduced per unit time}} = \frac{120}{\frac{11}{2}} = \frac{240}{11} = 21\frac{9}{11}$ minutes. Therefore, at $21\frac{9}{11}$ minutes past 4:00, the two hands will be together.

Ex. 30 Riya starts a trip when the hands of the clock are together between 8 am and 9 am. He arrives at his destination between 2 pm and 3 pm when the hands are exactly 180° apart. How long did the trip take?

Sol: At 8:00 am, the distance between the minute hand and the hour hand is 240°. It takes the minute hand $\frac{240}{\frac{11}{2}} = \frac{480}{11} = 43\frac{7}{11}$ min to catch up with the hour hand. Therefore, at $43\frac{7}{11}$ minutes past 8:00, the two hands are together. At 2:00 pm, the distance between minute hand and hour hand is 60°. Therefore, first the minute hand has to cover this gap of 60° and then get ahead by further 180° to make an angle of 180° with the hour hand. Therefore, at $43\frac{7}{11}$ minutes past 2:00, the two hands are 180° apart. Therefore, the total time taken for the trip = time taken from $8:43\frac{7}{11}$ to $2:43\frac{7}{11} = 6\text{hours}$.